SynAthlete-AIoT

An Explainable AIoT Framework for Athlete Injury-Risk Prediction Using GAN-Synthesized Wearable Data

Problem-Based Learning · **Sports-Focused, Software-Only AIoT Pipeline**

PROBLEM STATEMENT

Injury risk is invisible until it's too late

Coaches and athletes lack a real-time, explainable way to see injury and fatigue risk building up over time. Two structural gaps in current research and data make this hard to solve:

Design without implementation

Recent AIoT athlete-health frameworks propose wearable + edge-fog-cloud architectures, but explicitly leave anomaly detection, injury-risk prediction and explainability as unimplemented future work.

Snapshots, not timelines

Existing injury-risk ML gets strong accuracy, but only on cross-sectional, one-time data — not continuous, day-by-day wearable measurements that show risk building up.

A black-box bottleneck

State-of-the-art deep models for wearables are accurate but opaque and too heavy for edge hardware — so explainability at the edge remains an open problem.

THE CORE GAP

No dataset joins continuous vitals with real injury outcomes.

Physiological time-series datasets exist. Injury-outcome datasets exist. They have never been combined — leaving no ready-made way to train or validate a longitudinal, explainable injury-risk model.

LITERATURE SURVEY

What the field has built — and left open

1

MDPI APPLIED SCIENCES · MAY 2026

Conceptual AIoT Athlete-Health Framework

Proposes wearable sensing + edge-fog-cloud + multimodal analytics. Its own limitations section: the AI layer (anomaly detection, injury-risk, explainability) is design-only — never implemented.

2

BMC SPORTS SCIENCE · JAN 2026

SHAP + Random Forest Injury-Risk ML

Reaches 0.98 accuracy on injury-risk classification, but flags its own dataset as cross-sectional — no daily time-series like a wearable would provide.

3

IOT TIME-SERIES SYNTHESIS

TimeGAN / DoppelGANger Lineage

Established generative architectures for multivariate time-series from distributed IoT sensors — the citable lineage behind this project's synthetic-data module.

4

EDGE-AI HEALTH SURVEYS

Fragmented, Cloud-Dependent Solutions

Wearable-AI research tends to isolate single concerns (energy, anomaly detection, privacy) and depends on cloud compute — introducing latency, privacy and connectivity risk.

OBJECTIVES

What SynAthlete-AIoT sets out to do

1. **Bridge the data gap**

   Generate realistic, longitudinal synthetic athlete vitals with TimeGAN / WGAN-GP, conditioned on real injury-risk factors — joining physiological time-series with injury outcomes for the first time.

2. **Simulate tiered edge-fog-cloud processing**

   Build a software-only, latency-tiered pipeline (real-time alert → session aggregation → longitudinal modeling) — no physical hardware required.

3. **Predict injury risk with a fused model**

   Combine a time-series arm (LSTM/GRU/TCN) with a tabular arm (Random Forest) into one decision-level ensemble risk score.

4. **Make every risk flag explainable**

   Attach SHAP and attention-based explanations to each prediction, and validate them against known biomechanical risk factors — not just spurious correlations.

5. **Deliver a coach-facing dashboard**

   Surface live risk scores, trend charts and plain-language explanations so the output is usable, not just accurate.

DATASET

Two real, verified halves

PHYSIOLOGICAL / MOTION TIME-SERIES

NO INJURY LABELS

PAMAP2

9 subjects · wrist, chest & ankle IMUs + HR monitor · 18 activities @ 100Hz · ~3.85M timestamped instances

WESAD

15 subjects · chest & wrist wearables · ECG, EDA, EMG, respiration, temperature, 3-axis accel · stress-state labels

INJURY-OUTCOME DATA

NO CONTINUOUS TIME-SERIES

SIRP-600

Sports Injury Risk Prediction dataset — structured risk factors and outcomes

Injury Prediction (Runners)

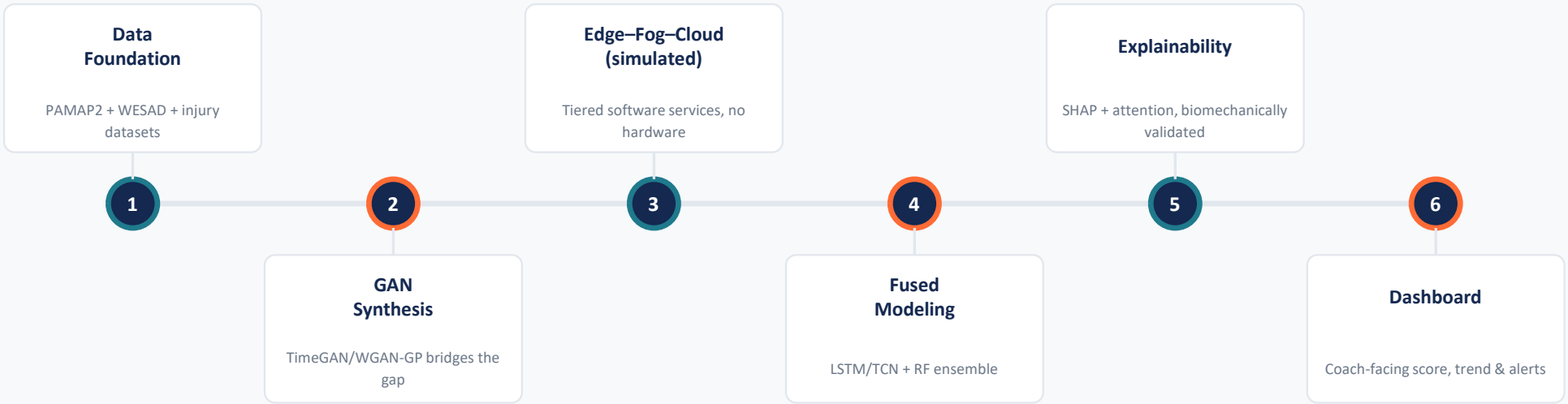
Competitive-runner injury records with training-load variables

Collective Sports Sensor DB

168 real cardiorespiratory sessions from real wearable sensors

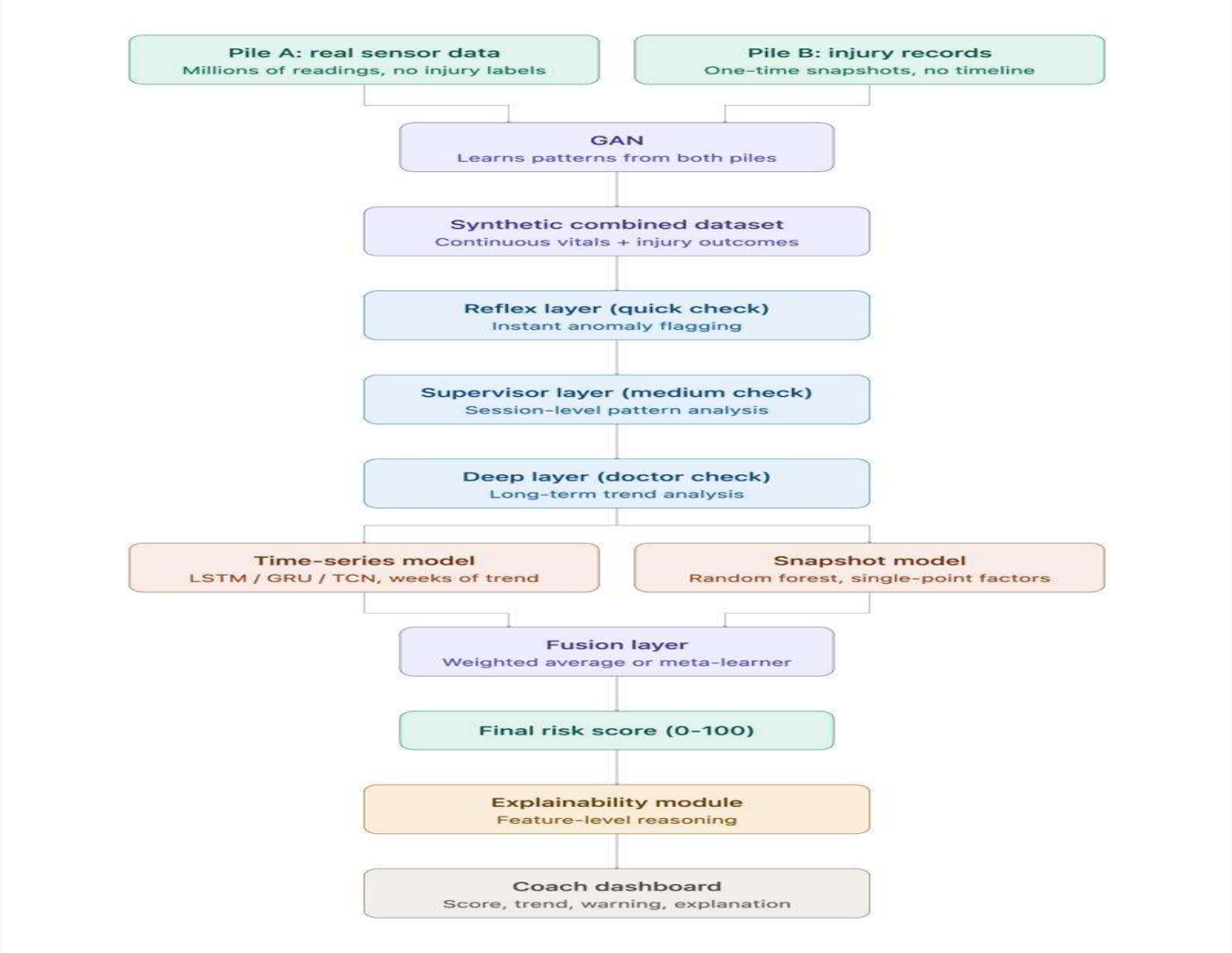
PROPOSED PIPELINE

From two real datasets to an explainable risk score



Software-only: every tier runs as a container/service on one machine — the tiered architecture is demonstrated without buying or wiring any physical sensor.

Architecture Diagram



EXPECTED CONTRIBUTION

Novelty claim

1 Implements the missing AI layer

Builds the anomaly-detection, injury-risk and explainability layer the May-2026 MDPI framework explicitly left as future work.

2 Snapshot → timeline

Extends cross-sectional injury classification (BMC, Jan 2026) into continuous, longitudinal risk trajectories.

3 Bridges fragmented data

GAN joins two real, disconnected dataset types instead of requiring an impossible all-in-one dataset.

4 Explainability that's validated, not assumed

Checks model attributions against known biomechanical risk factors — a methodological step most papers skip.